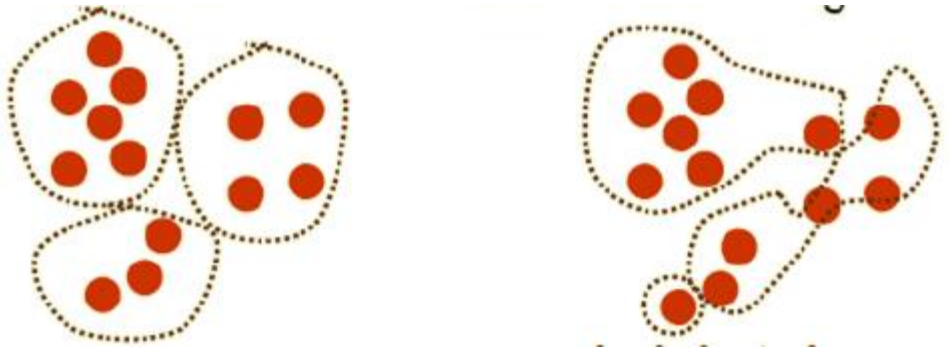
Measuring Quality of Clusters

Learning Objectives

- Interpret **cluster quality metrics** such as:
  - Silhouette Coefficient
  - Davies–Bouldin Index – self study
  - Dunn Index – self study
  - Adjusted Rand Index (ARI) and Mutual Information – self study

1. Introduction

- Consider the following two clusters



  - *“Which clustering looks better? Why?”*
  - We want to **quantify** with cluster quality metrics.

2. Why Cluster Quality Matters

- Clustering is **unsupervised**, so there are **no labels** to compare to — hence, we need ways to measure how “good” the clustering is.

2. Why Cluster Quality Matters

- Clustering is **unsupervised**, so there are **no labels** to compare to — hence, we need ways to measure how "good" the clustering is.

- Two main perspectives:
  - **Internal**: How compact and well-separated the clusters are.
  - **External**: How well the clusters match known labels (if available).
  - **Relative**: Comparing results from different algorithms or parameters.

```
# --- Import required libraries ---
import numpy as np
import matplotlib.pyplot as plt
from sklearn.metrics import silhouette_samples,
silhouette_score

from sklearn.cluster import KMeans

# --- Step 1: Create simple 2D dataset ---
X = np.array([
    [1, 2], [2, 2], [2, 3],   # Cluster 1
    [8, 8], [9, 8]            # Cluster 2
])
```

```
# --- Step 2: Apply KMeans clustering ---
kmeans = KMeans(n_clusters=2, random_state=42)

labels = kmeans.fit_predict(X)
```

```
# --- Step 3: Compute silhouette values ---
silhouette_avg = silhouette_score(X, labels)

sample_silhouette_values = silhouette_samples(X, labels)

print("Cluster labels:", labels)

print("Silhouette Coefficients for each point:",

np.round(sample_silhouette_values, 3))

print("Average Silhouette Score:", round(silhouette_avg,3))
```

```
# --- Step 4: Visualize clusters and centroids ---
plt.figure(figsize=(7, 5))

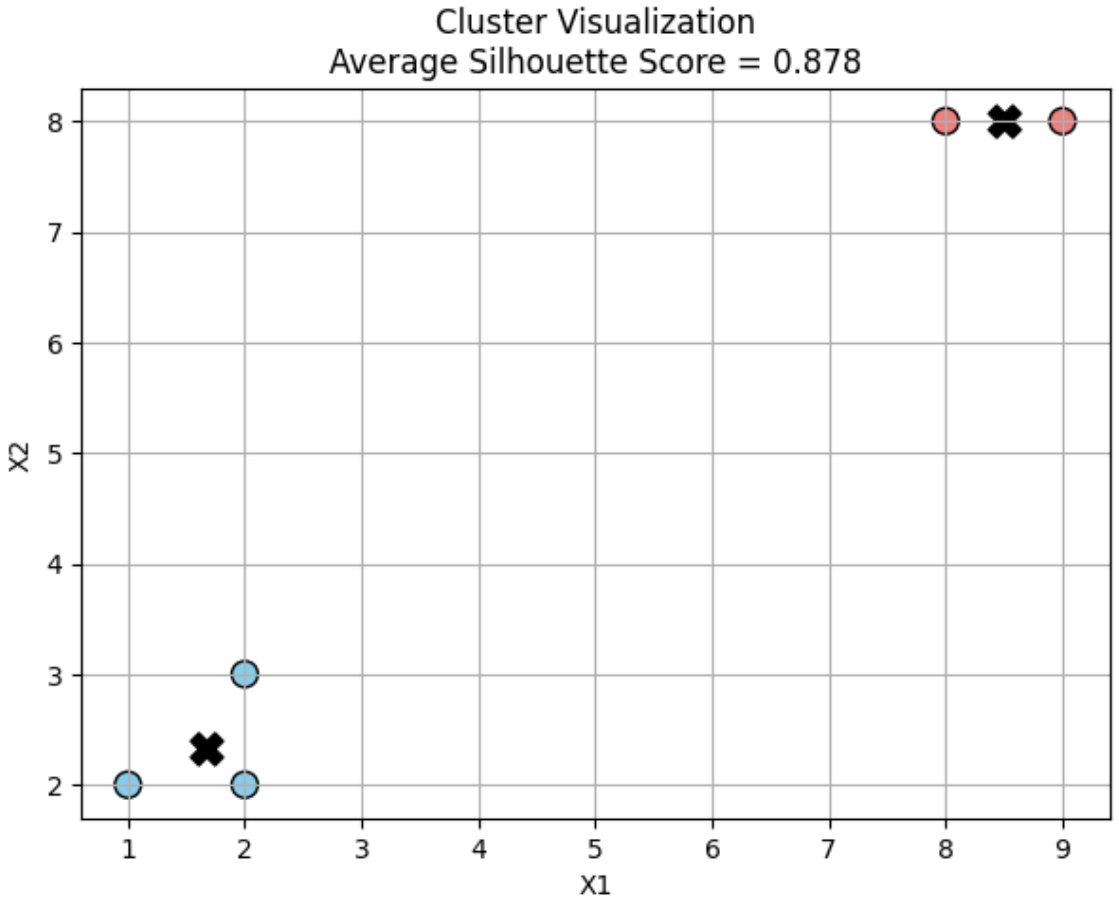
colors = ['skyblue' if label == 0 else 'lightcoral' for label
in labels]
plt.scatter(X[:, 0], X[:, 1], c=colors, s=100, edgecolor='k')
centroids = kmeans.cluster_centers_
plt.scatter(centroids[:, 0], centroids[:, 1], c='black',
s=150, marker='X')

plt.title(f"Cluster Visualization\nAverage Silhouette Score =

{silhouette_avg:.3f}")

plt.xlabel("X1")
plt.ylabel("X2")
plt.grid(True)
plt.show()
```

```
Cluster labels: [0 0 0 1 1]
Silhouette Coefficients for each point: [0.874 0.887 0.853
0.882 0.892]
Average Silhouette Score: 0.878
```



Average Silhouette Score = 0.877 → Excellent cluster quality
Both clusters are compact (points close to each other) and well-separated (far from the other cluster).

Internal Evaluation Metrics - Silhouette Score

- **Internal**: How compact and well-separated the clusters are.

- Formula:

$$sc = 1 - \frac{a}{b}$$

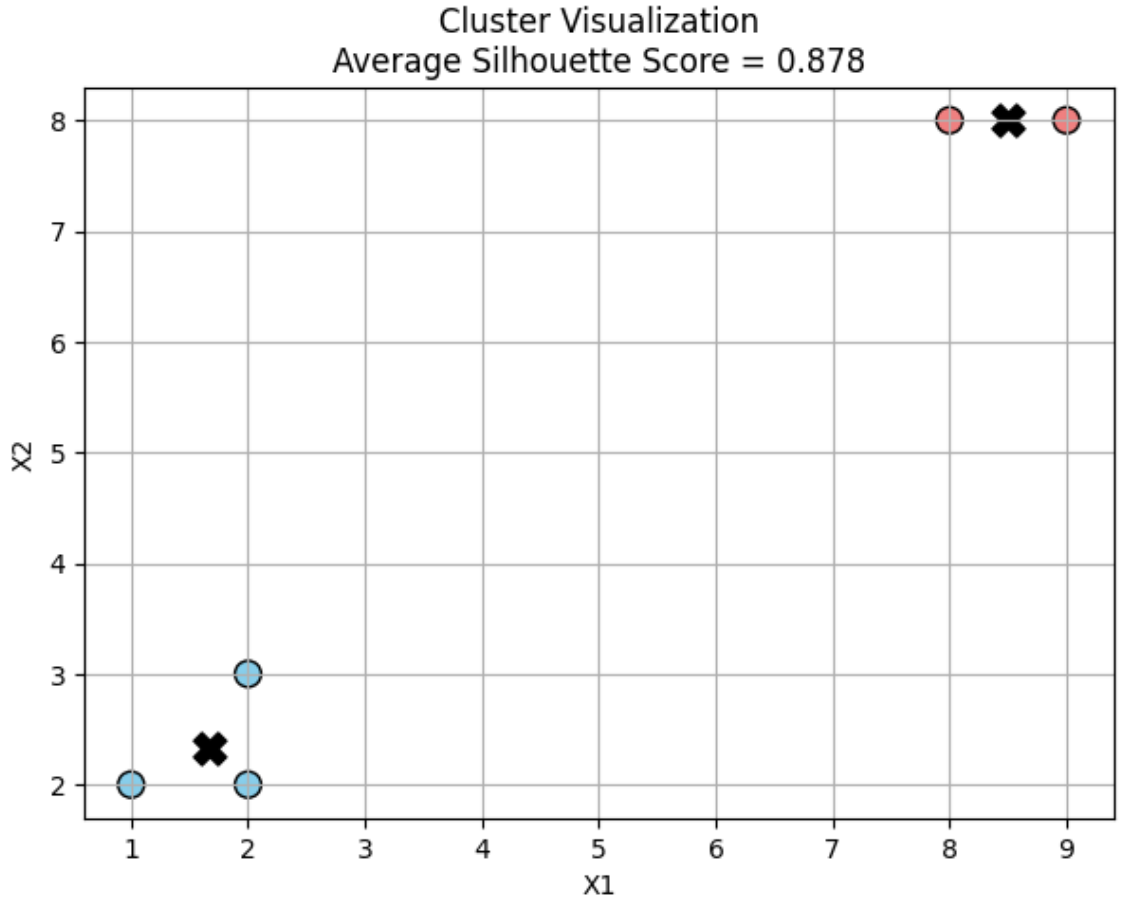
a - average distance within cluster

b - average distance to nearest other cluster

- Range: $[-1, 1]$;closer to 1 → better clustering.

Silhouette Score example

Point	Cluster	Coordinates (x, y)
A	1	(1, 2)
B	1	(2, 2)
C	1	(2, 3)
D	2	(8, 8)
E	2	(9, 8)



Thank you